YouTube Intro to Ranking Systems

Business Intelligence - DAT-8564 - B.STEM2

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Process Documentation

Hello future analysts! The steps below describe the entire process from data ingestion and cleaning, to handling inconsistencies, calculating DAX measures and finally designing the global ranking feature used to identify the top-5 trending videos.

1. **ETL Process Summary**

Importing the CSV Data

After downloading the JSON file “MX_category_id” and the CSV file “MXvideos”, open Power BI and in the Home Tab click “Get Data” and CSV file. Browse, select the file and click “open”. When the preview of your data appears, check the box and click “Transform Data” to clean the data. You are now in Power Query, click “New Source” then “More” and select JSON file, follow the same steps as before. You will have two (2) tables with each file name.

Let’s work on the CSV file “MXvideos” first. First we changed the source setting by changing the line breaks from “Apply all line breaks” to “ignore quoted line breaks” which allowed us to clean the rows that were blank and had insufficient and inconsistent data. Additionally, 49 rows were removed by selecting all the columns then clicking on “Remove Rows” and then “Remove Duplicates”. These steps are important because blank, insufficient or duplicated data can result in misleading insights when users interpret visualizations, or can cause steps to break later in the process.

Changing Column Titles

The column titles were in the first row so we selected “Use First Rows as Headers”. The following columns should be text (ABC): video_id, title, channel_title, tags, thumbnail_link and description. The following should be whole numbers (123): category_id, views, likes, dislikes and comment_count. The following should be True/False: comments_disabled, ratings_disabled and video_error_or_removed.

Adjusting the Trending Date

Let’s continue with the dates, which are the most tricky data type, so be very careful. The column trending_date is categorized as a text. As a result, if we try to change the type to date immediately, Power BI won't know how to interpret it because of inconsistent formats and regional format differences. Therefore, we need to right-click on trending_date, select “Split Column” and click “By Delimiter". The delimiter should be Custom “.” (Power BI would do it automatically). This will split the single column into three (3) columns, one representing day, one representing month, and one representing year. Let’s rename each column as “year”, “date” and “month” to avoid confusion, and change the data type to a whole number.

Before we merge these columns to create a single date column, we have to address the year column since PowerBI will not read “17” as “2017” or “18” as “2018”. Therefore, we used a DAX formula to customize the year column and add 2000 to each occurrence in the “year” column, because we need to have the year with four (4) digits as the scientific format requests. This generated a new column which contained “2017” and “2018”. We delete the old year column and bring the new year column next to “day” and “month”. Select the columns in the respective order of year, month and day, and then merge the columns with the custom separator “-”. Finally, change the data type to date and the format will be displayed according to your computer’s settings.

Adjusting the Publish Date

Now, let’s go to publish_date. This column is automatically a text type and is already in the scientific format. Therefore, we change data type to Date/Time/Timezone and then because we are only interested in the date, change it to Date and click “replace current”.

Remember to save your work, apply later and then click on close and apply, every time you have done a significant amount of steps.

Importing the JSON Data

Now let’s clean the JSON file “MX_category_id”. All columns were deleted except “items.id” and “items.snippet.title”. This was done to maintain only relevant data in our model.

The ID column should be a whole number (123) and the category title, a text (ABC). Additionally, the category title column had two “comedy” rows, which were removed by selecting the column and then removing duplicates. This is important because it would have created a many-to-many relationship (which we don’t want) between the two tables when we try to link the tables later.

All columns were renamed to clean and standardize the data for readability.

Finally, save and close and apply.

2. DAX

Creating a Calendar Table

Once we had our both tables cleaned, a calendar table was created by selecting the Modeling Tab and “Create New Table”. This table is a must in every business intelligence tool because it allows us to make better calculations and enables consistent filtering across the entire model.

Managing Table Relationships

Now, we have three (3) tables in our model, therefore, relationships must be done to ensure that they are connected and can interact with one another to make the visuals. We therefore created the following relationships: Many to one between CategoryID from the CSV (MXvideos) with CategoryID from the JSON file (MX_categoryID) because the CSV file has multiple rows each with a category ID and the JSON file has one unique row per category ID, and many to one between TrendingDate from the CSV (MXvideos) and Date from the new Calendar table because the CSV file has multiple entries with the same date whereas Calendar has one unique row per date.

Performing Calculations for Analysis

Now, let's proceed with the challenging but interesting part of the process, creating calculations that enable deeper data analysis and support better decision-making. The first step is to analyze what we want our output to be. We need to model a ranking system that allows the user to select a date and a category, so that the user will be able to see the top-5 trending videos on that date and within that category.

To create our ranking system, we are given four (4) metrics: views, likes, dislikes and comments. We ensured that our top-5 trending videos were dynamic by creating measures for each metric. Measures also work better than columns with filters, making them more reliable for the analysis.
Our data contains videos that may or may not have the comments disabled (boolean column in the CSV file). As a result, we opted to exclude comments as an engagement metric.
Our ranking system is therefore determined by views (popularity), likes (approval) and dislikes (penalty). However, you are right to think that an engagement metric is crucial to determine if people are actually interested in that type of content. Therefore, we included an engagement rate by combining likes and dislikes, because it shows that users are not only passively watching the video.

We must then normalize the four (4) metrics so that no single metric overpowers the rest. This step is important because views are significantly larger than likes and dislikes, and if a difference in scale exists when designing the ranking system, the metric with the largest magnitude would dominate the overall rank score. Therefore, we used min-max normalization to adjust the views metric to a range between 0 to 1, where 0 represents the lowest views and 1 the highest views. This step assures us that the popularity rate is comparable in terms of scale with the approval rate, penalty rate and engagement rate.

After we created our popularity rate measure, we created the other rates using measures to ensure that the data is not static like columns. We calculated approval rate with "likes/(likes+dislikes)" , penalty rate with "dislikes/(dislikes+likes)" and engagement rate with "(likes + dislikes)/views". Additionally, users need to be able to visualize the number of days a video has been trending. This new measure counts the number of distinct dates on which the video appeared in the trending dataset up to the selected date. Other measures like % of Likes, % of Dislikes, average engagement and approval rate were created to enhance user understanding of the top-5 videos based on selected category and date.

Creating the Algorithm

Now, we can proceed to create the algorithm for the Ranking Score of the videos. A measure was created by assigning weights to each of our metrics. Popularity rates were given a weight of 35% since views are the main driver of trending content in every social media platform. Engagement was given a weight of 30% since it determines whether the video was sufficiently engaging to capture the user interest. Approval rate was given a weight of 20% to indicate that the video had a positive impact on the user, while the penalty rate was given a weight of 15%. We believe that the penalty rate is also important to determine trending videos because although it expresses negative feedback, it still drives visibility and interaction. The weights are then multiplied with their corresponding rate and added together to obtain the final "Ranking Score". As a result, we end up with a value between 0 and 1 which is used to determine the top trending videos for specific categories on a specific date.

3. Dashboard

Now, let's continue with the creative part of this job: designing the visuals. Be aware because designing visuals is not only about aesthetics, but more so about what is important for the audience to analyze, in this case what is important for the viewers to see from trending videos.

Firstly, to display the trending videos, we chose a Table and included the columns for the thumbnail, the video title, number of views, number of comments, % of Likes, % of Dislikes, number of days trending and ranking score. We also want to display the video thumbnail, so we added the ThumbnailLink as an image. To do so, click in the column and in the Data Category, select "Image URL".

Our goal is to present the top-5 trending videos per date and category, so two (2) slicers were then added in the dashboard. A slicer was added for the Date (which is our trending date) and the other for Category Title. The date slicer was placed first, so that the user would select a date, after which they would select a category. This ensures that the visuals react in the order we intend.

Up to this point, we have a table containing all videos categorized by date and category but we only need the top-5 videos. To achieve this, the column "Title" is filtered in the table using the "Top N" filter type which allows us to display the top 5 videos by the Ranking Score value.

Additional KPIs, such as total views and average engagement and approval rate based only on the trending videos were included to improve the user's data analysis. To ensure transparency, the visualization highlights the metrics considered in the ranking methodology.

ESR/ESG Discussion

The dilemma between media content visibility and freedom of speech is growing as more users are harmed by brand's media choices, and lawsuits continue to be filed (Escamilla, 2026). Importantly, Jacobson et al (2020) state that ethical responsibilities extend beyond legal requirements.

In March 2026, the New Mexico jury ordered Meta to pay $375 million in damages for failing to warn users about platform design features that endangered children to sexual predators (Kerr, 2026). Attorney Torrez identified this as the price for Meta's choice to put profits over kids' safety (New Mexico Department of Justice, 2026). The verdict pressures Meta to redesign its features and rethink its media strategy to avoid further litigation and losing users to reputational damage.

On the other hand, Unilever made a market-driven decision in 2020 to stop buying Facebook ads since it believed that violent and divisive speech was not being effectively managed on the platform (Feiner, 2020). Six years later, Unilever's brand safety policy prohibits advertisement in any media known for promoting violence, pornography or insulting behaviour (Unilever, 2026). Unilever holds to its belief that companies are responsible for creating a trusted and safe digital ecosystem (Feiner, 2020 ; Adams, 2020), but this could result in loss of visibility.

Similarly, YouTube reached a reputational turning point when major advertisers like Coca-Cola and Amazon pulled ads from the platform after finding their content paired with hate speech and violence (Maheshwari, 2017). In response, YouTube became stricter with its advertising policies and announced that it was leading advertisers toward "content that meets a higher level of brand safety" (Maheshwari, 2017). In contrast, Disney was court-mandated to adjust its YouTube strategy after being found guilty for violating the Children's Online Privacy Protection Act (*Disney Agrees to $10M Civil Penalty*, 2025). The tangible consequence was $10 million in civil penalties, while the implicit damage was consumer distrust.

Thirty years have passed since Section 230 was created, but technology has evolved far beyond its original context (Harvard Gazette, 2026). While both legal and market constraints are flawed, commercial behavior remains shaped by financial incentives before ethical considerations, meaning that significant change is more likely to happen if it affects revenue. However, platforms' main focus will always be to increase engagement because it drives profits, even at the risk of user dissatisfaction and harm. For instance, our algorithm provided a positive weighting to video dislikes, acknowledging that a highly-disliked video can still be trending. Although this is statistically accurate, it is not ethically defensible. The responsibility of regulating and screening content should therefore be borne by both the content creator and the platform.

Market constraints allow free speech content and prioritize transparency for the viewer. Therefore, it represents the better option for platform regulation. Consumer behavior has also shifted, particularly among younger generations that have grown in a digital environment and expect sustainability and ethical company practices. Ultimately, outdated laws cannot continue to protect social media platforms. There should be clear, high-standard accountability and consequences for online speech and visibility.

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